

A trace has been found.

Abbreviations
$\sim M_1 = \text{TLS12}$
$\sim M_2 = \text{rs_4}$
$\sim M_3 = \text{DHE_cipher}$
$\sim M_4 = \text{group_G}$
$\sim M_5 = \text{sign}(\text{spk}(\text{sskS_2}), \text{caSKey_1})$
$\sim M_6 = \text{group_G}$
$\sim M_7 = g$
$\sim M_8 = \text{exp}(g, y_3)$
$\sim M_9 = \text{sign}((\text{group_G}, g, \text{exp}(g, y_3), a, \text{rs_4}), \text{sskS_2})$
$\sim M_10 = \text{ServerHelloDone}$
$\sim X_1 = (\text{exp}(g, a_2), \text{CCS}, \text{PRF}(\text{PRF}(\text{exp}(\sim M_8, a_2), \text{master_secret}, (a, \sim M_2)), \text{client_finished}, \text{Hash}(((\text{TLS12}, a, \text{DHE_cipher}), (\text{TLS12}, \sim M_2, \text{DHE_cipher}, \text{group_G}), (\text{group_G}, g, \sim M_8, \sim M_9), \text{exp}(g, a_2))))))$ $= (\text{exp}(g, a_2), \text{CCS}, \text{PRF}(\text{PRF}(\text{exp}(\text{exp}(g, a_2), y_3), \text{master_secret}, (a, \text{rs_4})), \text{client_finished}, \text{Hash}(((\text{TLS12}, a, \text{DHE_cipher}), (\text{TLS12}, \text{rs_4}, \text{DHE_cipher}, \text{group_G}), (\text{group_G}, g, \text{exp}(g, y_3), \text{sign}((\text{group_G}, g, \text{exp}(g, y_3), a, \text{rs_4}), \text{sskS_2})), \text{exp}(g, a_2))))))$

